

Sales Forecasting & Demand Intelligence System

Executive Business Report | Prepared for the Head of Supply Chain and the CFO

Executive Summary

This report summarizes what four years of company sales data (2015–2018) tell us about future demand, and what the business should do about it. We built and tested three different forecasting methods, checked the sales history for unusual spikes and drops, and grouped our products into demand-based segments so that stocking decisions can be tailored rather than one-size-fits-all. The most accurate model predicts monthly sales to within roughly 15% of the actual figure. The data shows a clear, repeating holiday surge every November–December, a small group of products (Phones, Chairs, Storage, Tables, Binders, Accessories) that reliably drive the bulk of revenue, and the West region together with the Furniture category showing the strongest growth over the next quarter. The sections below turn these findings into specific forecast numbers, risk flags, and stocking actions.

Key Findings from Data Analysis & Forecasting

- The dataset covers 9,800 orders placed between January 2015 and December 2018, across three product categories, 17 sub-categories, and four regions. The data was clean: no duplicate records and no meaningful missing values.
- Sales follow a genuine, statistically-confirmed pattern: an underlying upward trend, plus a strong seasonal cycle that repeats every year (confirmed using a standard statistical stationarity test). This means the swings in monthly sales are not random noise — they are predictable enough to forecast.
- The clearest example of this seasonality: November 2018 sales hit \$117,938, a roughly 52% jump from October 2018 (\$77,448) — consistent with holiday-season buying.
- Three forecasting approaches were built and compared head-to-head on historical data:

Forecasting Model	Typical Error (USD)	Typical % Off from Actual
SARIMA (statistical)	\$19,244	20.5%
Prophet (industry tool)	\$20,251	21.9%
XGBoost (machine learning)	\$15,398	15.0%

The machine-learning model (XGBoost) was the most accurate of the three and is recommended as the primary forecasting engine going forward, refreshed monthly as new sales data comes in.

3-Month Sales Forecast (in Plain Language)

Below is the forecast from our statistical model, which is the one that also produces a realistic best-case / worst-case range rather than a single number:

Month	Expected Sales	Reasonable Range
January 2019	~\$46,800	\$17,000 – \$76,600
February 2019	~\$40,300	\$9,600 – \$71,000
March 2019	~\$72,200	\$41,300 – \$103,100

In plain terms: we expect a dip in February before sales pick back up in March. The most accurate model (XGBoost) agrees with this shape — roughly \$53,000 in January, dipping to \$33,500 in February, then rebounding to \$60,200 in March. The wide ranges above are intentional and honest: with only four years of history to learn from, a single exact number would be misleading. Use the midpoint for planning and the range for risk assessment (e.g., worst-case stock levels).

Top 3 Anomalies Detected & Likely Cause

Week Of	Sales	Likely Cause
March 22, 2015	\$37,704	Sudden spike (5x+ a typical week) — consistent with one or more large bulk orders landing the same week.
December 2, 2018	\$35,999	End-of-year / holiday promotional buying.

Week Of	Sales	Likely Cause
November 18, 2018	\$30,572	Black Friday / holiday-season shopping surge.

11 unusual weeks were flagged in total (both high spikes and unusually quiet weeks, the quiet ones mostly falling right after the holidays). A second, simpler cross-check method flagged none of these same weeks, which tells us the primary detection method is currently the more sensitive one and the one worth trusting for alerts — but the disagreement itself is a signal the detection logic should be reviewed periodically (see Risk & Limitations).

Product Demand Segmentation & Recommended Stocking Strategy

Segment	Products	Profile	Recommended Stocking
High Volume, Stable Demand	Phones, Chairs, Storage, Tables, Binders, Accessories	Largest combined sales (\$239K), steady ~18% growth, moderate order value (~\$361)	Keep standard, well-buffered inventory — these are the safe, predictable sellers.
Declining / Low Volume	Bookcases, Appliances, Furnishings, Paper, Supplies, Art, Envelopes, Labels, Fasteners	Smaller combined sales (\$54K), lowest average order value (~\$143)	Keep lean stock, order in smaller/more frequent batches, avoid tying up capital in slow movers.
Growing, High-Value	Machines, Copiers	Smaller volume but highest average order value (~\$1,931) and most volatile sales pattern	Avoid large standing stock; use just-in-time or pre-order replenishment, but line up supplier capacity for growth.

Business Recommendations

- 1. Prioritize the West region and the Furniture / Office Supplies categories for the next quarter's stocking budget.** Our per-category, per-region forecast shows the West region growing the most over the next three months (+\$12,683), followed by Furniture (+\$10,163) and Office Supplies (+\$6,636), while Technology is projected to decline slightly (-\$2,494). Shift incremental stock and marketing spend accordingly.
- 2. Build a seasonal buffer for November–December.** Historical data shows a 50%+ month-over-month sales jump going into the holidays (Nov 2018 vs Oct 2018) and repeated unusually high sales weeks clustered around Black Friday and year-end. Plan extra safety stock, warehouse space, and staffing about a month ahead of the season, rather than reacting to the spike after it starts.
- 3. Adopt the XGBoost model as the primary forecasting engine and tier stocking by demand segment.** It produced the lowest error of the three models tested (~15% vs. 20–22% for the alternatives). Pair this with segment-based stocking — standard buffers for the High Volume/Stable segment, lean batches for the Declining segment, and just-in-time ordering for the Growing/High-Value segment — to free up working capital currently tied up in slow-moving inventory.

Risk & Limitation

This system was trained on only 48 months of monthly sales history — a small sample for statistical forecasting. That is why the confidence ranges above are wide (for example, the January estimate could reasonably fall anywhere between \$17,000 and \$76,600). These forecasts should be used as directional guidance for planning conversations, not as guaranteed figures, and the models should be recalibrated every quarter as new sales data becomes available. The anomaly-detection system should also be reviewed periodically, since its two underlying methods did not fully agree on this dataset.